ArgusSAR: A Multidimensional Perception and Robust Detection Method for Multiscale Objects in SAR Images

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***Abstract*—Due to the challenges presented by the different scales of objects and the complex interference in synthetic aperture radar (SAR) images, current detectors struggle to achieve satisfactory accuracy and exhibit weak generalization abilities across different datasets. This article proposes a robust and universal object detector framework named ArgusSAR, which is suitable for both ship and aircraft object detection. The proposed framework comprises three key components. In detail, a spatial encoder (SE), which employs multihead attention (MHA) blocks integrated with multilayered encoder blocks, extracts robust feature representations while suppressing irrelevant interference inherent in complex SAR backgrounds. Moreover, a fine-grained feature aggregation encoder, which incorporates the multilevel pyramid architecture with content-aware upsampling operations, enables precise integration of multidimensional features and enhanced extraction of object boundaries. Furthermore, a robust multidimensional awareness attention decoder (MAAD), which incorporates scale–spatial–task self-attention mechanisms to strengthen geometric perception and improve both localization accuracy and generalization ability for ship and aircraft objects. Three publicly available SAR datasets involving ship and aircraft objects are chosen to confirm the effectiveness of ArgusSAR. Compared with the baseline and the other methods, our method has the best performance on these datasets. It demonstrates that the ArgusSAR framework is robust for SAR ship and aircraft object detection. The source code is available at https://github.com/masurq/ArgusSAR**

***Index Terms*—Deep learning, multiscale, multitask, object detection, synthetic aperture radar (SAR).**

I. Introduction

SYNTHETIC aperture radar (SAR), owing to its special imaging mode, not only has the ability to observe the earth at a high resolution widely and periodically, but also has all-day and all-weather characteristics [1], [2]. Furthermore, owing to the significant differences in the backscatter coefficients between metal materials and the background in SAR images, man-made objects can be effectively distinguished. Therefore, it is the most effective tool for accurately detecting and monitoring typical objects, and has obtained widespread application in both military and civil areas [3].

In the initial period of object detection, the traditional detection methods are typically based on sea–land segmentation and constant false alarm rate (CFAR) detection [4]. However, these CFAR and its variants are designed for specific scenes and inevitably come with the disadvantages of a lower detection accuracy and longer detection time [5], [6], [7], [8]. Furthermore, these methods are significantly affected by noise interference. Besides, it is vital to be aware that traditional methods have difficulty obtaining ideal detection results when faced with massive amounts of data [9].

With the advancement of artificial intelligence technology and computer equipment, deep learning detection methods for SAR images have received widespread attention [10], [11]. Convolutional neural networks (CNNs) can automatically extract features of interesting objects, simplifying the object detection process and eliminating the dependence on human intervention [12], [13]. The common object detectors involve two-stage and one-stage strategies. Examples of the classical two-stage object detectors based on CNNs are R-CNN, Fast R-CNN [14], and Faster R-CNN [15]. These object detectors perform well in terms of detection accuracy. However, due to the need for region proposal, the speed of these detectors is slow. One-stage object detectors adopt the method of regression to identify the class and location of the objects directly. The classical one-stage object detectors based on CNNs are the single-shot detector (SSD) [16], [17], and the "You Only Look Once" (YOLO) series [18], [19], [20]. In detail, the YOLO-based algorithms perform well in detection speed and accuracy. These methods, originally applied in computer vision for natural images, also obtain an excellent performance in object detection. Due to similar spectral characteristics with natural images, optical images have been widely applied in object detection tasks based on these algorithms [21], [22], [23], [24].

However, there is an obvious difference in the scattering characteristics between SAR images (as Fig. 1) and optical images, which means that these algorithms are difficult to

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apply directly in the SAR field. Considering the unique mechanism of SAR backscattering, the main problems of detectors are as follows.

1) *Discontinuous Object Scattering and Structured Interference:* Unlike in optical images, objects such as ships and aircraft in SAR images often appear as a collection of discrete, strong scattering centers rather than as continuous contours. Simultaneously, the background contains speckle noise and structured interference from other man-made objects with high reflectivity, like bridges and port facilities. This requires the detector to suppress speckle noise and understand the object's structural information, linking discrete scattering points into a whole while distinguishing them from structured interference.
2) *Detail Information Loss for Multiscale Objects:* Small objects in SAR images often have low signal-to-noise ratios, and their detailed features are easily lost during sampling due to speckle noise or feature fusion processing. Moreover, man-made objects like ships and aircraft frequently exhibit sharp boundaries resulting from strong scattering effects. However, during multiscale feature fusion via upsampling, common context-agnostic interpolation methods (such as bilinear or nearest) inevitably smooth these sharp boundaries. This not only degrades the localization precision but can also lead to confusion of densely arranged small objects at the feature level, making them difficult to distinguish.
3) *Object Shape Diversity Due to Imaging Geometry:* Considering the imaging mechanism of SAR, variations in radar wavelength and viewing angle can lead to pronounced differences in the backscattering coefficients of the same object in the imagery. Moreover, even under identical imaging conditions, object rotation or differences in its surrounding environment may result in significant variations in the observed backscattering coefficients. This diversity introduced by the imaging mechanism demands that the detector's decoding module possess high robustness and adaptability to effectively cope with the complex and variable scattering characteristics.

Recently, detectors based on CNNs have significantly contributed to ship and aircraft detection in SAR images, which has solved the above difficulties to a certain extent. In detail, Zhang et al. [25] proposed the YOLO-FA method by introducing the frequency attention mechanism (FAM), which can enhance the quality of SAR images and emphasize the detection of ships above clutter. The conclusion shows that complex interference can be suppressed by introducing the attention module to focus on the interesting object. Furthermore, to address the issue of multiscale objects, scholars have made different optimizations to the algorithms. For example, Yang et al. [26] proposed the RFIM module, which uses multiple sizes of convolutional kernels for splicing and combines a CoAM module to obtain multiscale information. The experiments conducted in this study showed that this method outperformed the other methods by 2.6% in AP50 on the HRSID dataset. Wan et al. [27] proposed ESPHead to obtain the accurate features of objects by introducing convolutions with different dilation rates. Cui et al. [28] proposed a detector that combined the attention pyramid network with the CBAM module, which can accurately detect multiscale ships. The above detection methods introduce a series of convolution kernel sizes into the network and construct the feature extraction network with an attention module to accurately detect small objects. Moreover, to improve the robustness of the detector. Chen et al. [29] proposed CSD-YOLO with a spatial pyramid pooling module and a shuffle attention (SA) module, and introduced a novel loss function to improve the performance of the detector for different scales. This detection method achieved an excellent degree of performance in SAR images. Yu et al. [30] designed CR2A-Net by a rotated anchor-aided and align convolution module, and achieved a better detection performance. By improving the processing method of the fusion feature maps, the location and regression abilities of the above methods were improved, which in turn improves the generalization ability across different datasets. Shen et al. [31] proposed ELLK-Net, which significantly improves inference speed on edge devices while maintaining high detection accuracy through adaptive convolution kernel selection and a large-kernel multiscale attention module.

The improvement of aircraft detection, which aims to accurately locate and identify aircraft objects in a complex background, is more difficult than ship detection. Bao et al. [32] proposed an SAFP module, which was designed to accurately extract boundary feature information and combine it with a fine-grained attention mechanism to improve the precision of multiscale aircraft object detection. Huang et al. [33] introduced EST-YOLOv5s with the task-specific context decoupling head, which aims to balance the interaction between classification and regression. The results demonstrated that the EST-YOLOv5s detector achieved a recall (R) of 94.2%, a precision (P) of 97.3%, and an mAP of 97.8%. Guo et al. [34] combined scattering information enhancement and an attention pyramid network to effectively handle the separateness and variability of aircraft. Experiments verified the excellent performance, which obtained an AP of 83.25%. Zhang et al. [35] employed a multiscale fusion strategy to integrate the location information with semantic information, effectively realizing precise detection of SAR aircraft objects. The above methods obtain superior object detection results by constructing a fine-grained feature extraction module and improving the regression and classification performance of the detector for objects.

According to the above research, the YOLO-based methods have been verified to be able to effectively detect typical objects in SAR imagery with high speed and accuracy, especially ship and aircraft [36]. However, these methods underestimate information loss caused by the upsampling operation, and the traditional detection heads are unable to deal with different SAR objects. Aiming at the above-mentioned issues, we propose a robust detector named ArgusSAR for different ship and aircraft object detection, which is suitable for detecting multiscale ship and aircraft objects against complex backgrounds in SAR images. In detail, to address

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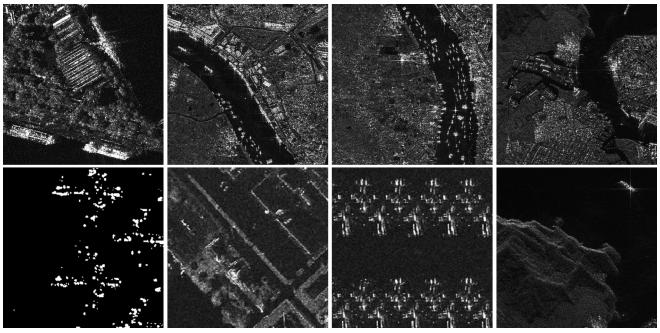


Fig. 1. Ships and aircrafts in SAR images.

the challenge of inadequate feature extraction in complex backgrounds, we develop a spatial encoder (SE). This encoder mitigates the impact of interfering objects in SAR images by combining local and global features. Furthermore, we design a fine-grained feature aggregation encoder (FFAE) to receive hierarchical features from the SE and perform multiscale feature fusion. This module generates feature maps with richer and more detailed spatial–textural details and semantic features. In addition, to resolve the instability of the detector when handling multiclass and multiscale objects in SAR images, a novel multidimensional awareness attention decoder (MAAD) is proposed, which can cross feature layers for scale–spatial–task awareness.

The major contributions presented in this article are as follows.

1) ArgusSAR detector, involving multilevel feature maps, fine-grained upsampling operator, and multidimensional attention modules, has a state-of-the-art (SOTA) performance compared to other latest methods in both ship and aircraft datasets of multiscale SAR images.
2) In SE, multihead attention (MHA) can not only adjust the weights of different channels to enhance the representation of object features, but also reduces the computation of unnecessary information, suppressing the irrelevant features of other objects in the background.
3) In FFAE, the content-aware upsampling operation enhances fine-grained features and reduces information loss during the feature map transmission process, allowing the network to be more sensitive in capturing the details of objects, demonstrating that the accuracy detail information can improve detection results.
4) MAAD can combine the self-attention mechanism with the output channels of multiscale and multitask perception, which significantly enhances the robustness of the detector.

II. Methodology

In Fig. 2, the ArgusSAR framework is composed of SE, fine-grained feature aggregation encoder, and MAAD.

The SE effectively addresses the challenge of robust feature extraction in SAR images characterized by densely arranged objects and strong interference. This encoder operates by integrating a multilayer convolutional encoding module, designed for hierarchical feature extraction, with an MHA. The multilayer convolutional module enables efficient multilevel feature extraction, and the MHA dynamically adjusts the weights of different feature channels and regions. It can concentrate on discriminative features of objects, such as ships and aircraft, while effectively suppressing background clutter and interference signals.

Moreover, in low-resolution SAR images, ship and aircraft objects often appear as small pixels, typically performing as fine scattering points or edge features. They are highly sensitive to the effects of noise and strong reflections from various ground objects. To address the poor performance of traditional upsampling methods in small object detection, we propose an FFAE. This module employs a multilevel pyramid structure to integrate feature representations from different stages. Specifically, this module uses learnable kernels for content-aware upsampling based on the SAR image content, replacing conventional interpolation methods. This innovative approach allows for the precise reconstruction of high-resolution feature maps containing object boundary information and fine-grained details, while simultaneously preserving high-dimensional semantic features. This leads to more accurate feature fusion of different scales and ultimately improves the detection performance of small objects.

Besides, to overcome the inadequate localization capabilities and poor generalization of traditional detection heads when faced with multiclass object detection across diverse scales, orientations, and task requirements (e.g., classification and regression), we design a novel MAAD. This module combines scale-aware attention for adapting to different object sizes, spatial-aware attention that uses deformable convolutions to focus on distinct spatial locations, and task-aware attention. By optimizing feature channels to suit various object sizes, spatial positions, and task demands, this decoder enhances its perceptual capabilities regarding object scale and spatial location.

A. Spatial Encoder

In general, the ship and aircraft objects show different characteristics in SAR images. The multilayer encoder block is designed to extract different object features by deep and robust convolutional structure. However, when faced with densely arranged objects in SAR images, the CNN-based network is easily affected by the interfering objects, which obtains weak feature expression and results in false detection. Therefore, we combine the MHA block with the encoder block to create the SE, which can concentrate on important objects and suppress the interference of irrelevant features.

Specifically, the SE is composed of M encoder blocks that capture a variety of features within an image through multilayer convolution operations. Each encoder block not only extracts fundamental spatial characteristics but also gradually delves into more complex semantic levels. This process provides a robust foundation of information for subsequent feature fusion and object detection. Furthermore, to focus more on critical information, the SE incorporates the MHA block.

Since SAR imaging is dependent on the object's material composition and orientation relative to the radar beam,

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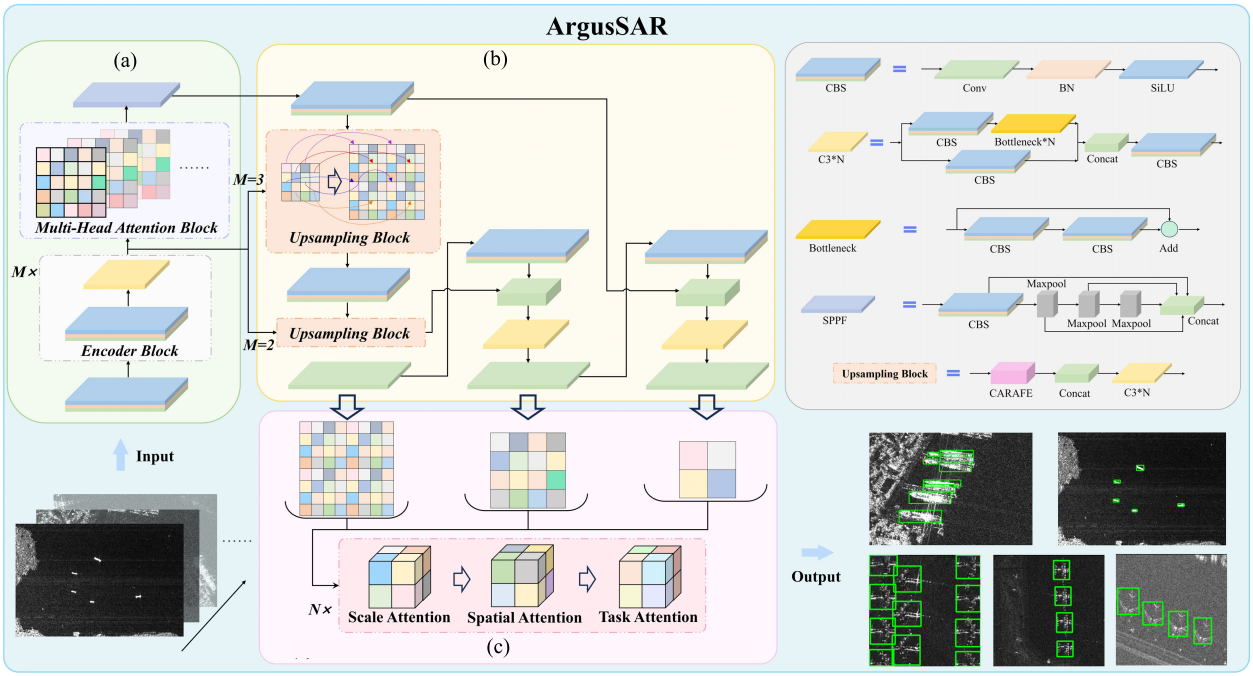


Fig. 2. Overall structure of the ArgusSAR network. (a) SE. (b) Fine-grained feature aggregation encoder. (c) MAAD.

the object's structural information and spatial context are crucial for accurate detection. MHA can capture long-range dependencies and connect semantically related regions that are distant from each other in an image. Specifically, it can semantically associate strong scattering centers that are discrete and spatially distant on the object (such as a ship's bow and stern). This process captures the structural entirety of the object, thereby effectively addressing the challenge of discontinuous strong scattering in SAR images. By learning global dependencies, the model can acquire more discriminative features between objects (e.g., ships) and interfering facilities (e.g., coastal buildings) that may share similar local scattering properties but differ in overall structure.

The workflow of the MHA block is shown in Fig. 3.

1) *Input Embedding:* The input features are represented as an embedding matrix $E \in \mathbb{R}^{n\times d}$, in which n and d correspond to the sequence length and embedding dimension, respectively.
2) *Linear Transformations:* For each attention head h, the query (Q), key (K), and value (V) matrices are derived from the input embedding matrix through three distinct linear transformations

$$Q_h = EW_{Q_h}, \quad K_h = EW_{K_h}, \quad V_h = EW_{V_h} \tag{1}$$

where W_{Q_h}, W_{K_h}, and W_{V_h} are the weight matrices associated with the hth attention head.
3) *Attention Score Calculation:* Within each attention head h, we scale the dot product of the query and key matrices by dividing by $\sqrt{d_k}$, and then apply the Softmax function to get the attention score matrix

$$A_h = \text{Softmax}\left(\frac{Q_h K_h^T}{\sqrt{d_k}}\right). \tag{2}$$

4) *Weighted Summation:* The attention score matrix A_h is used to weight the value matrix V_h, resulting in the output Z_h for the hth attention head

$$Z_h = A_h V_h. \tag{3}$$

5) *Concatenation and Transformation:* The concatenated outputs of all attention heads are projected through a final linear layer to produce the output matrix Z

$$Z = \text{Concat}\,(Z_1, Z_2, \ldots, Z_H)\, W_O \tag{4}$$

where H is the number of attention heads, and W_O is the final output weight matrix.

Thus, the MHA block enables the model to dynamically adjust the weights of different features. Owing to the MHA block, it effectively suppresses irrelevant features from interfering objects, ensuring that the model can concentrate on the characteristic representations of ship and aircraft objects.

B. FFAE

Pyramid structures, designed with top-down pathways and lateral connections, are effective at fusing semantic and localization information. By integrating feature information from different network depths, which cover varying receptive fields, the resulting fused feature maps inherently contain more diverse and comprehensive information than those from a single level. Therefore, we implement multilevel feature fusion based on a pyramid structure. It leverages this informational richness to address challenges in SAR images presented by factors such as variations in viewing angles, geometric distortions, and speckle noise.

However, the commonly used feature fusion pyramid structure is based on PANet, which uses the nearest neighbor upsampling strategy. It only pays attention to the neighborhood

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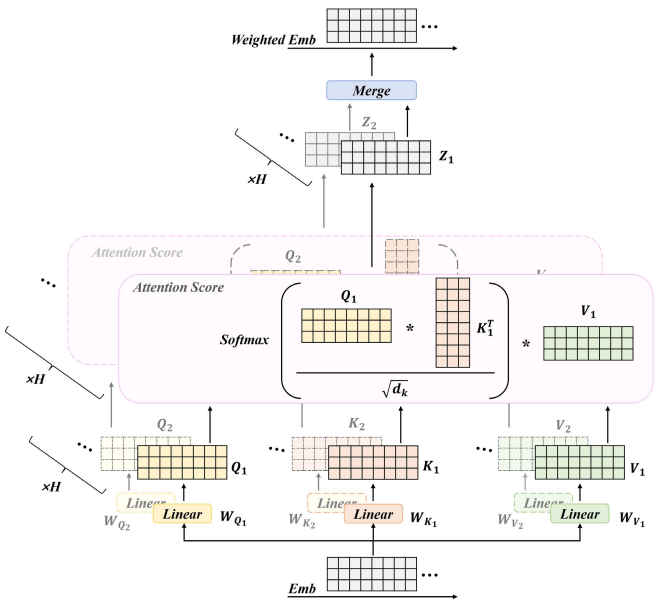


Fig. 3. Structure of the MHA block.

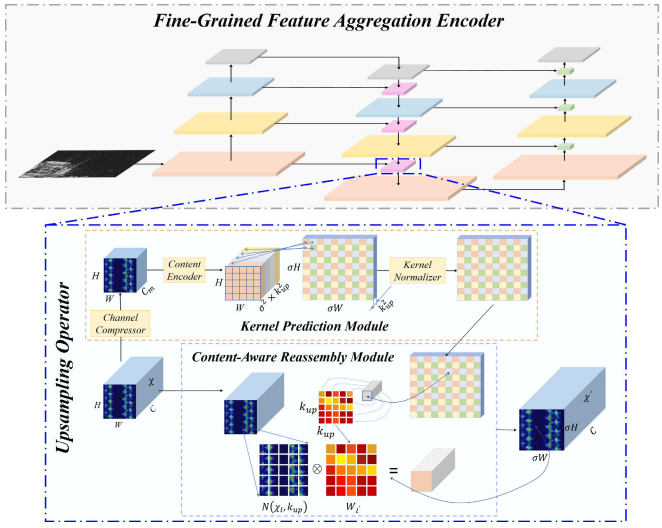


Fig. 4. Structure of the FFAE.

of the sample point in SAR images. Therefore, it fails to use semantic information, and hard to extract boundary information of the SAR objects precisely.

To address the limitations of traditional interpolation methods, which can blur distinct SAR object boundaries due to strong scattering and lose information on small objects, we design the FFAE module. As shown in Fig. 4, the FFAE adopts a multilevel pyramid structure. This hierarchical design integrates multiscale feature information, where features from lower levels provide rich spatial details, while features from higher levels capture more abstract semantic information. A core component of FFAE is a content-aware and adaptive upsampling operator [37], which learns to predict reassembly kernels based on local features. This operator is designed to more effectively preserve the critical boundary structures and fine-grained details specific to SAR objects, and generate higher quality feature maps for subsequent fusion. By employing this refined multiscale feature interaction and fusion mechanism, FFAE can precisely combine information from different stages of the SE and generate richer and more detailed representations of spatial texture and semantic features.

In the pyramid architecture, the upsampling path can progressively restore low-resolution, high-level features to high resolution by using an upsampling operator, while simultaneously fusing them with low-level features. Through its content-aware upsampling method, the FFAE can account for the geometric shapes and scattering characteristics of the object, enabling a more accurate reconstruction of the boundaries and detailed information of objects.

Algorithm 1 Content-Aware Upsampling Operator of FFAE

Input: Input feature map $X \in \mathbb{R}^{C\times H\times W}$, upsampling factor σ, channel compression channels C_m, encoder kernel size k_{encoder}, reconstruction kernel size k_{up}

Output: Upsampled feature map $Y \in \mathbb{R}^{C\times \sigma H\times \sigma W}$

Step 1: Kernel Prediction

$X' \leftarrow \text{Conv1}\times 1(X, \text{out_channels} = C_m)$

$K_{\text{flat}} \leftarrow \text{Conv}(X', \text{kernel_size} = k_{\text{encoder}}, \text{out_channels} = \sigma^2 k_{\text{up}}^2)$ ▷Predict kernel weights, $K_{\text{flat}} \in \mathbb{R}^{H\times W\times(\sigma^2 k_{\text{up}}^2)}$

$K_{\text{pred}} \leftarrow \text{PixelShuffle}(K_{\text{flat}}, \sigma)$ ▷$K_{\text{pred}} \in \mathbb{R}^{\sigma H\times \sigma W\times k_{\text{up}}^2}$

$K_{\text{norm}} \leftarrow \text{SpatialSoftmax}(K_{\text{pred}})$ ▷$K_{\text{norm}} \in \mathbb{R}^{\sigma H\times \sigma W\times k_{\text{up}}^2}$

Step 2: Content-Aware Reconstruction

Initialize $Y \in \mathbb{R}^{C\times \sigma H\times \sigma W}$

for each object position (i, j) **do**

  Calculate corresponding $i = [i'/\sigma], j = [j'/\sigma]$

  $N \leftarrow \text{Neighborhood}(X, \text{center} = (i', j'), \text{size} = k_{\text{up}})$

  $W_{ij} \leftarrow \text{Reshape}(K_{\text{norm}}[i, j, :], (k_{\text{up}}, k_{\text{up}}))$

  $Y[:, i, j] \leftarrow \sum_{p=0}^{k_{\text{up}}-1} \sum_{q=0}^{k_{\text{up}}-1} W_{ij}[p, q] \cdot N[p, q, :]$

end for

return Upsampled feature map Y

This approach ensures that high-resolution features capture fine-grained texture details, while low-resolution features preserve rich semantic information. The upsampling operator is divided into a kernel prediction module and a content-aware reconstruction module. The kernel prediction module dynamically and adaptively predicts the optimal kernel weights for high-resolution feature map reconstruction based on local feature content. This ensures the upsampling process with content-awareness and learnability. The content-aware reconstruction module applies these specifically predicted kernels to each position of the low-resolution image to output the feature at the corresponding location in the predicted high-resolution feature map. This mechanism, founded on content-adaptive prediction and feature reconstruction, can effectively restore fine-grained textures and structural details, generating upsampling results that are far superior to those achieved with traditional, fixed interpolation methods. Algorithm 1 shows the upsampling process.

It is assumed that the upsampling ratio is σ and the input has a shape of $C \times H \times W$, which means the shape of the output is $C \times \sigma H \times \sigma W$. The process of the upsampling process is described as follows.

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1) *Kernel Prediction Module:* This module dynamically generates specific upsampling kernels for the subsequent reconstruction process by input feature map. First, the input feature X' is processed by $k_{encoder}$ size of convolutional layer. The size of $k_{encoder}$ determines the range of the local contextual receptive field that the module can use when predicting the upsampling kernels, and the number of output channel is $\sigma^2 k_{up}^2$. Next, these predicted weights are spatially unfolded to generate a corresponding upsampling kernel size of $k_{up} \times k_{up}$ for each position in the high-resolution map. Finally, each generated upsampling kernel is independently normalized using a Softmax function, obtaining the final normalized upsampling kernels that guide the content-aware reconstruction module.

2) *Content-Aware Reconstruction Module:* This module receives the original low-resolution feature map as well as the upsampling kernels generated by the kernel prediction module for each high-resolution output position as its input. For each output position, this module first locates and extracts a local neighborhood of size $k_{up} \times k_{up}$ from its corresponding location in the low-resolution input feature map. Next, it retrieves the specific upsampling kernel that was predicted for this object position. Finally, it uses this upsampling kernel to perform a weighted sum of the features within the extracted low-resolution neighborhood. The result of this operation is the high-resolution fused feature map.

To compute each pixel in the output high-resolution feature map, it is necessary to extract a local neighborhood from the input low-resolution feature map as the basis for the calculation. Therefore, the size of this neighborhood directly determines the richness of the local contextual information that the model can use from the low-resolution map when generating the high-resolution features. To accurately predict a large-sized (k_{up}) upsampling kernel that can effectively guide feature reconstruction, the content encoder requires a sufficiently large receptive field ($k_{encoder}$) to capture adequate local context. Consequently, the size of k_{up} must be aligned with $k_{encoder}$ to achieve optimal performance. In our subsequent experiments, we validate the optimal parameters for ship and aircraft object detection in SAR images.

C. Multidimensional Awareness Attention Decoder

Traditional algorithms often employ simple regression strategies to localize objects in SAR images. However, when processing SAR image that contains multiple objects, this approach often fails to achieve precise and reliable detection. These failures are typically due to factors such as feature confusion between objects and variations in scattering characteristics with changing viewing angles. To overcome these limitations, it is crucial to design a robust and universal decoder for SAR object detection, which should possess the following characteristics to address the unique challenges of SAR images [38]. In detail, SAR images often cover vast areas, resulting in significant variations in size. The detector should be scale-aware, capable of adaptively processing features at different scales. This ensures that small objects are not overlooked while the structural integrity of large objects is maintained. Furthermore, the spatial location of objects exhibits diversity and uncertainty due to imaging geometry, orientation, and the presence of complex background clutter. Therefore, the decoder should be spatial-aware, enabling it to precisely focus on the critical spatial regions of the object. Besides, SAR object detection can involve diverse representation requirements; the decoder also needs to be task-aware. This allows it to optimize feature representations for specific prediction tasks, such as allocating appropriate feature channels for different objectives, thereby enhancing the synergy and accuracy of multitask processing.

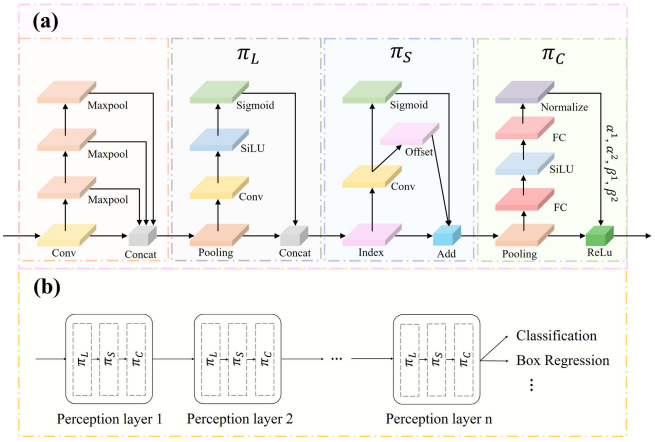


Fig. 5. Structure of (a) MAAD and (b) apply to detector.

To address the above challenges, we design the MAAD. Its core idea is to build a universal decoder with strong adaptability to SAR object diversity by integrating multiple self-attention mechanisms. In Fig. 5(a), before entering the attention modules, the input feature map first passes through a parallel, multiscale max-pooling layer to generate a noise-robust feature map. The feature map is then fed into π_L, π_S, and π_C self-attention mechanisms. In Fig. 5(b), these attention mechanisms are combined as a perception layer.

Algorithm 2 shows the workflow of MAAD. The detailed introductions to the functions of each module are as follows.

1) *Maxpool Component:* Within the MAAD module, the initial Maxpool component constructs a spatial pyramid structure through parallel multiscale pooling layers, serving a dual critical role in processing SAR images. In detail, by selecting the maximum backscatter value within local regions across different scales, it effectively enhances the strong scattering signals of objects, thereby improving their salience against background clutter and speckle noise. Furthermore, this structure forms a spatial pyramid pooling, enabling the module to capture contextual information at different scales and providing more comprehensive information for the subsequent attention mechanisms.

2) *π_L (Scale-Aware Attention):* The pixel dimensions of the same object class can vary by an order of magnitude. For example, since the size of samples in the HRSID dataset differs from the SADD dataset, standard convolutional receptive fields struggle to effectively capture context for objects at such different scales. π_L mechanism addresses this by performing global average pooling on the multiscale pooled features to

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Algorithm 2 Workflow of MAAD Module

Input: Input feature map $F \in \mathbb{R}^{L\times H\times W\times C}$ from FFAE
Output: Refined feature map $F_{out} \in \mathbb{R}^{L\times H\times W\times C}$ for final prediction

Step 1: Multiscale Context Feature Extraction
$P_1 \leftarrow \text{MaxPool}(F, \text{kernel_size} = k_1)$
$P_2 \leftarrow \text{MaxPool}(F, \text{kernel_size} = k_2)$
$P_3 \leftarrow \text{MaxPool}(F, \text{kernel_size} = k_3)$
$F_{\text{pooled}} \leftarrow \text{Concat}(P_1, P_2, P_3)$ ▷Suppress speckle noise and build feature pyramid
Step 2: Scale-aware Attention (π_L)
$v \leftarrow \text{GlobalAvgPool}(F_{\text{pooled}})$ ▷Vector $v \in \mathbb{R}^{L\times 1\times 1\times C'}$
$A_L \leftarrow \sigma(\text{Linear}(v))$ ▷Scale attention weights $A_L \in \mathbb{R}^{L\times 1\times 1\times C}$
$F_L \leftarrow A_L \otimes F$ ▷Apply weights via channel-wise multiplication
Step 3: Spatial-aware Attention (π_S)
$\Delta p, \Delta m \leftarrow \text{Conv}(F_L)$ ▷Learn offsets Δp and modulators Δm
$F_S \leftarrow \text{DeformConv}(F_L, \Delta p, \Delta m)$ ▷Flexible sampling grid focuses on key spatial regions
Step 4: Task-aware Attention (π_C)
$\theta \leftarrow \text{HyperNet}(F_S)$ ▷Generate params $\theta = \{\alpha^1, \beta^1, \alpha^2, \beta^2\}$
$F_{out} \leftarrow \max(\alpha^1 \otimes F_S + \beta^1, \alpha^2 \otimes F_S + \beta^2)$ ▷Dynamically gate channels for tasks
return Refined feature map F_{out}

generate a channel vector describing the global scene context. This vector is then passed through a small fully connected layer to output a set of weights, A_L, with the same number of channels as the original feature map. This process is equivalent to understanding the context of the entire feature map to decide which channels to enhance. For instance, if the scene contains numerous fine-grained features, it will boost channels sensitive to small objects. Conversely, it will enhance channels sensitive to the structure of large objects.

3) π_S *(Spatial-Aware Attention):* Due to the side-looking geometry characteristic of SAR images, backscattering of ship hull or an aircraft is affected by observation angle and surrounding environment. This variability causes standard convolution to inevitably include irrelevant background pixels, which can contaminate the extracted features. The core of π_S is deformable convolution, which learns an additional 2-D offset, Δp, for the standard convolutional grid. This means the sampling points are no longer confined to a fixed grid but can dynamically adjust their positions according to the object's shape, forming a flexible receptive field that tightly conforms to the object's contour. For example, when detecting a slender ship, the sampling points can align along the ship's main axis, thereby maximally extracting effective features while ignoring background interference on either side.

4) π_C *(Task-Aware Attention):* In SAR images, features for localization and classification are sometimes separated. For instance, a few discrete strong scattering points may be sufficient for the model to classify "there is an object here," but precisely regressing its bounding box requires information from the object's overall contour. π_C addresses this issue via a HyperNet. It allows the model to autonomously decide whether to improve the response of strong scattering points (beneficial for classification) or to preserve and enhance edge contour information (beneficial for localization), thus dynamically optimizing the feature flow for different tasks.

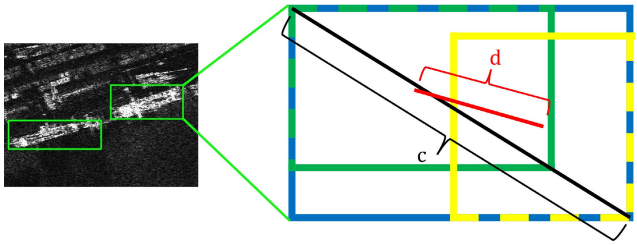


Fig. 6. CIoU loss.

D. Loss

For the proposed approach, the loss function is composed of localization $Loss_{CIOU}$, confidence $Loss_{conf}$, and classification $Loss_{class}$, which is defined as follows:

$$Loss = Loss_{CIOU} + Loss_{conf} + Loss_{class}. \tag{5}$$

In Fig. 6, the yellow box, red box, and blue box represent the predicted bounding box, the ground-truth box, and the minimum enclosing rectangle, respectively. The complete intersection over union (CIOU) loss [39] is defined as follows:

$$CIOU = IOU - \frac{\rho^2\left(b, b^{gt}\right)}{c^2} - \alpha v \tag{6}$$

where $\rho^2(b, b^{gt})$ represents the distance between the centers of the ground-truth box and the predicted bounding box, and c represents the diagonal length of the minimum enclosing rectangle. The parameters α and v are as follows:

$$\alpha = \frac{v}{1 - IOU + v} \tag{7}$$

$$v = \frac{4}{\pi}\left(\arctan\frac{\omega^{gt}}{h^{gt}} - \arctan\frac{\omega}{h}\right). \tag{8}$$

$Loss_{CIOU}$ is defined as follows:

$$Loss_{CIOU} = 1 - IOU + \frac{\rho^2\left(b, b^{gt}\right)}{c^2} + \alpha v. \tag{9}$$

$Loss_{conf}$ represents the position regression loss, which is defined as follows:

$$\begin{aligned}\text{Loss}_{conf} &= \sum_{i=0}^{S^2}\sum_{j=0}^{B} I_{ij}^{obj}\left[\hat{C}_i^j \log\left(C_i^j\right) + \left(1 - C_i^j\right)\log\left(\hat{C}_i^j\right)\right] \\ &\quad - \lambda_{noobj}\sum_{i=0}^{S^2}\sum_{j=0}^{B} I_{ij}^{noobj}\left[\hat{C}_i^j \log\left(C_i^j\right) + \left(1 - C_i^j\right)\log\left(\hat{C}_i^j\right)\right]\end{aligned} \tag{10}$$

where I_{ij}^{obj} and I_{ij}^{noobj} represent whether an object, λ_{noobj} represents the counterbalance of the loss caused by the negative samples, $\hat{C}_i^j$ represents the predicted bounding box, C_i^j

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represents the ground-truth box, and S indicates that the row and column number of the picture are both S.

$Loss_{class}$ represents the object class loss, which is defined as follows:

$$Loss_{class} = \sum_{i=0}^{S^2} I_{ij}^{obj} \left[\hat{P}_i^j \log\left(P_i^j\right) + \left(1 - \hat{P}_i^j\right) \log\left(1 - P_i^j\right) \right] \tag{11}$$

where P_i^j represents the label, and $\hat{P}_i^j$ represents the prediction score.

III. Experiments

A. Datasets and Evaluation Metrics

1) Dataset Description: The detection performance of ArgusSAR was evaluated on the SAR datasets of SSDD [47], HRSID [48], and SADD [35]. These datasets contain various SAR ship and aircraft objects with different scales, making them suitable for evaluating the effectiveness of different object detectors.

a) SSDD: The 1160 images containing 2456 ship objects are from different SAR sensors. The ship objects are made up of white pixels, while the land and sea are mostly composed of black pixels. Every SAR image contains a distinct variety of ship objects, varying in number, size, and category. Although this dataset has fewer images than other datasets, it is specifically labeled for only ship objects, making it more effective when training ship detectors. What is more, the partitioning of the training set and test set is 8:2.

b) HRSID: The 136 panoramic SAR images are from different SAR sensors. The images are cropped to 5604 SAR images with dimensions of 800×800, and the resolution varies between 0.5 and 3 m. There are 16 951 ships in the dataset. In this dataset, 40% of the images were randomly selected according to the proportion of nearshore and offshore images in the original dataset, and the partitioning of the training set and test set is based on the configuration file.

c) SADD: The SADD dataset is collected by the TerraSAR-X satellite. The ground-truth box of the aircraft in these images has been manually annotated based on the real optical images and expert knowledge. About 2966 nonoverlapped images with a size of 224×224 were collected with 7835 aircraft objects after cropping the initial large images. The partitioning of the training set and test set is divided into a 5:1 ratio.

2) Experimental Settings: All the experiments are conducted with Python 3.9.18, PyTorch 2.3.1, CUDA 12.1, and an NVIDIA A800 80 G GPU. To handle the differences in dataset sizes and ensure consistency in experiments, we scale all input images to 512×512 pixels during both training and testing stages. The mosaic method is used to augment the input images. During training, the SGD optimizer is used to update the network parameters iteratively. The momentum learning rate and weight decay are set to 0.937 and 5×10^{-4}, respectively. The model is trained for a maximum of 500 epochs, and the training stops early if the performance metrics no longer improve. We employ the k-means clustering strategy to generate the prior knowledge of bounding boxes and determine the configurations of these boxes across different datasets. Besides, during the training process, the results of the last epoch and the best epoch are both saved.

3) Evaluation Metrics: The metrics, including P, R, $F1$-score, and mAP@.5, are chosen to evaluate ArgusSAR's performance. Metrics are calculated as follows:

$$P = \frac{TP}{TP + FP} \tag{12}$$

$$R = \frac{TP}{TP + FN} \tag{13}$$

$$mAP@.5 = \int_0^1 P(R)\, dR \tag{14}$$

$$F1 = \frac{2 \times P \times R}{P + R}. \tag{15}$$

B. Comparative Experiments

1) Comparison With Baseline Methods: We conducted experiments on SSDD, HRSID, and SADD, three datasets, and compared ArgusSAR with baseline methods. The results are listed in Table I.

a) SSDD: The SSDD dataset includes numerous inshore ships, where coastlines and port facilities create strong, structured interference. On this dataset, ArgusSAR shows the best detection performance, compared with YOLOv8, with increases of 1.6%, 2.0%, and 1.5% in P, R, and mAP@.5, respectively. ArgusSAR's ability to simultaneously improve both P and R metrics over YOLOv8 indicates that our model can more accurately identify objects from complex backgrounds (attributed to the SE module) and also detect more easily overlooked small objects (attributed to the FFAE module). These improvements enable the proposed detector to perform better in ship localization and classification tasks. Moreover, when compared with YOLOv10, the proposed method demonstrates a 1.0% advantage in $F1$-score, showing its capability to precisely detect ship objects against complex environmental backgrounds.

b) HRSID: The core challenge of the HRSID dataset lies in the high density of small, closely arranged ship objects set against complex backgrounds. In this specific scenario, the $F1$-score is a more revealing metric as it demands a strict balance between P and R. The proposed method benefits from the content-aware upsampling operator to reconstruct ship feature maps. It helps the detector distinguish the differences between ship objects and land objects, thus achieving an $F1$-score of 91.4%. When compared with YOLOv8 and YOLOv10, the proposed method outperforms them by 1.9% and 0.4%, respectively, while effectively balancing the P and R metrics. The superior $F1$-score indicates our model achieves a high R metric without sacrificing precision by adding more background false alarms. We attribute this primarily to the content-aware upsampling operator at the core of the FFAE module. This operator reconstructs ship feature maps with greater fidelity, helping the detector better distinguish ship objects from land clutter and enabling excellent performance in both P and R metrics.

c) SADD: The SADD dataset contains complex airport backgrounds—such as runways, aprons, and buildings—that

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TABLE I

COMPARISON OF THE PERFORMANCE OF DIFFERENT OBJECT DETECTION METHODS

Method	SSDD (Ship Dataset)				HRSID (Ship Dataset)				SADD (Aircraft Dataset)			
	P (%)	R (%)	F1 (%)	mAP@.5 (%)	P (%)	R (%)	F1 (%)	mAP@.5 (%)	P (%)	R (%)	F1 (%)	mAP@.5 (%)
SSD [17]	90.3±1.52	84.7±1.75	87.4±1.16	88.4±2.26	87.4±3.61	79.3±2.89	83.2±2.30	88.8±3.58	82.8±1.33	90.4±1.59	86.4±1.04	91.7±1.87
YOLOv3 [40]	91.7±1.26	85.8±1.81	88.7±1.11	92.3±1.95	83.4±3.19	84.1±2.38	83.7±1.98	84.7±3.34	87.9±1.11	94.5±1.48	91.1±0.93	95.7±1.95
YOLOv4 [41]	94.8±1.23	90.8±1.59	92.8±1.01	95.4±1.33	90.2±2.85	84.5±2.11	87.3±1.76	90.5±2.92	91.8±0.95	94.9±1.20	93.3±0.77	97.1±1.44
YOLOv5 [42]	93.2±0.76	92.3±0.88	92.7±0.58	96.2±0.78	91.5±2.16	86.1±1.87	88.7±1.42	91.2±2.25	95.7±0.79	97.1±0.94	96.4±0.62	97.7±1.35
YOLOv6 [43]	92.4±1.47	94.7±1.22	93.5±0.95	96.7±1.15	92.1±2.33	87.4±1.72	89.7±1.44	92.8±2.46	95.4±0.81	97.5±1.00	96.4±0.65	97.9±1.26
YOLOv7 [44]	96.7±0.99	93.1±0.91	94.9±0.67	96.9±1.03	89.9±2.96	76.6±3.52	82.7±2.31	84.8±3.27	96.1±1.28	97.4±1.10	96.7±0.84	98.1±1.62
YOLOv8	96.2±0.61	95.4±0.54	95.8±0.41	97.1±0.79	**93.7**±1.65	85.6±1.78	89.5±1.22	92.8±1.99	94.2±1.43	**98.9**±1.86	96.5±1.19	**98.6**±2.03
YOLOv9 [45]	95.7±0.86	96.2±0.95	95.9±0.64	97.4±0.91	93.2±1.83	86.7±2.01	89.8±1.37	92.6±2.44	96.4±0.96	97.8±0.89	**97.1**±0.65	98.2±1.24
YOLOv10 [46]	**96.9**±0.72	**96.4**±0.59	**96.6**±0.46	**98.2**±0.81	92.9±1.77	**89.1**±2.38	**91.0**±1.50	**93.1**±2.06	**96.9**±1.25	97.1±0.94	97.0±0.77	98.4±1.39
YOLOX [19]	93.9±0.84	91.7±0.92	92.8±0.62	95.7±1.07	92.2±2.42	84.6±1.91	88.2±1.53	92.4±2.63	91.6±0.80	96.7±1.03	94.1±0.66	97.3±1.24
ArgusSAR	**97.8**±0.55	**97.4**±0.37	**97.6**±0.33	**98.6**±0.43	**93.5**±1.36	**89.4**±1.55	**91.4**±1.48	**93.1**±1.60	**97.1**±0.84	**98.6**±0.77	**97.8**±0.57	**98.9**±1.11

* The gray background represents our method **ArgusSAR**. Numbers in red indicate the best performance, while numbers in blue indicate the second-best performance.

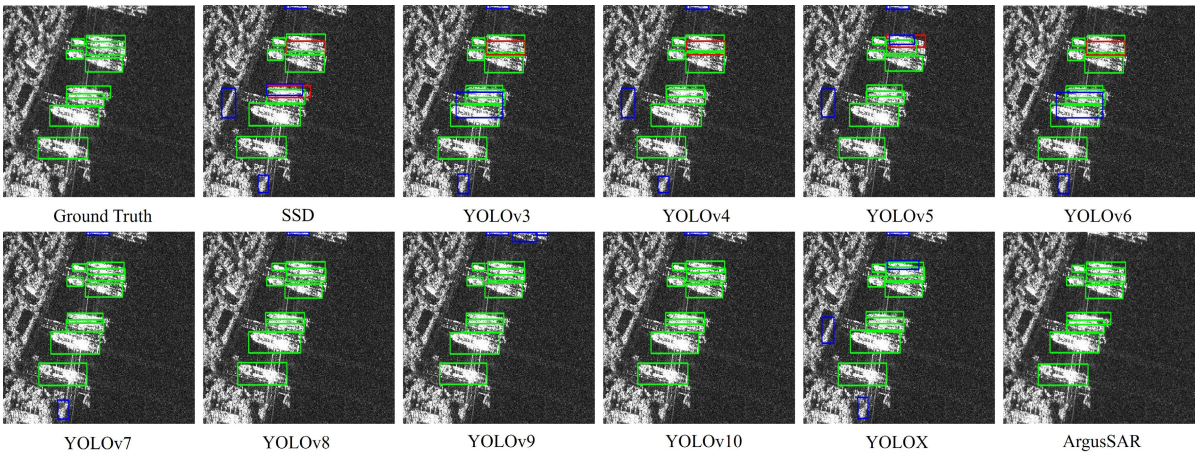


Fig. 7. Detection results on the SSDD dataset. In these images, the green boxes, blue boxes, and red boxes represent correct detections, false detections, and missed detections, respectively.

are easily confused with aircraft objects. ArgusSAR significantly outperforms the YOLOv10 method by 0.8% in $F1$-score and by 0.5% in mAP@.5, showing a considerable improvement. When compared with YOLOv8 and YOLOX methods, ArgusSAR achieves 97.8% of P metric, which is superior to these comparison methods. This significant increase in precision demonstrates that our model (particularly the SE and MAAD modules) has a distinct advantage in distinguishing objects from structured background interference. It also confirms that ArgusSAR maintains excellent performance for detecting small aircraft under complex interference conditions and is suited for SAR-based aircraft detection tasks.

In summary, it has been shown that ArgusSAR can achieve an excellent performance on different ship and aircraft datasets and can meet the practical application requirements.

2) Visual Results: To particularly exhibit the enhanced performance of ArgusSAR, it is essential to visualize the detection results across the above three SAR datasets.

Figs. 7–9 present visualized results of ArgusSAR compared to the other methods. In detail, Figs. 7–9 show the visualized results on the SSDD, HRSID, and SADD datasets, respectively. It can be clearly observed that ArgusSAR performs better than any other methods in correct detections. In addition, ArgusSAR is also effective in detecting densely arranged ships, and the detection results are shown in Fig. 10. Furthermore, from Fig. 11, ArgusSAR also demonstrates a superior performance in detecting small ship objects. The detection results for aircraft objects are shown in Fig. 12.

TABLE II

COMPARISON OF THE DIFFERENT SOTA OBJECT DETECTION METHODS ON THE SSDD DATASET

Method	P (%)	R (%)	F1 (%)	mAP@.5 (%)
YOLO-FA [25]	95.2	95.0	95.1	96.8
CSD-YOLO [28]	95.9	95.9	95.9	**98.6**
BL-Net [49]	91.3	96.1	93.6	96.8
LMSD-YOLO [50]	96.5	94.1	95.3	98.0
HR-SDNet [51]	96.5	91.0	93.7	90.8
DB-YOLO [52]	87.8	97.5	92.4	97.8
HyperLi-Net [53]	90.1	96.7	93.3	96.1
AFSar [54]	94.1	**98.2**	96.1	97.7
HA-SARSD [55]	97.0	92.0	94.4	97.0
ArgusSAR	**97.8**	97.4	**97.6**	**98.6**

3) Comparison With Other SOTA Methods: To evaluate the effectiveness of ArgusSAR, Tables II–IV provide a quantitative comparison between ArgusSAR and the other SOTA object detection methods on the SSDD, HRSID, and SADD datasets, respectively.

In Table II, ArgusSAR surpasses the AFSar method by 3.7% in precision and 0.9% in mAP@.5. The $F1$-score of ArgusSAR also exceeds that of AFSar by 1.5%. In addition, ArgusSAR shows a superior detection performance with the

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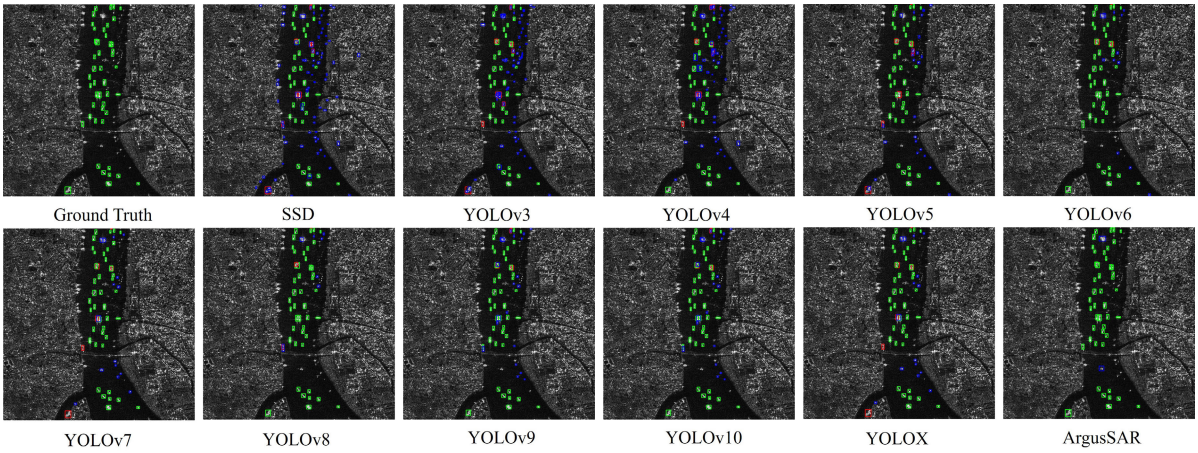


Fig. 8. Detection results on the HRSID dataset. In these images, the green boxes, blue boxes, and red boxes represent correct detections, false detections, and missed detections, respectively.

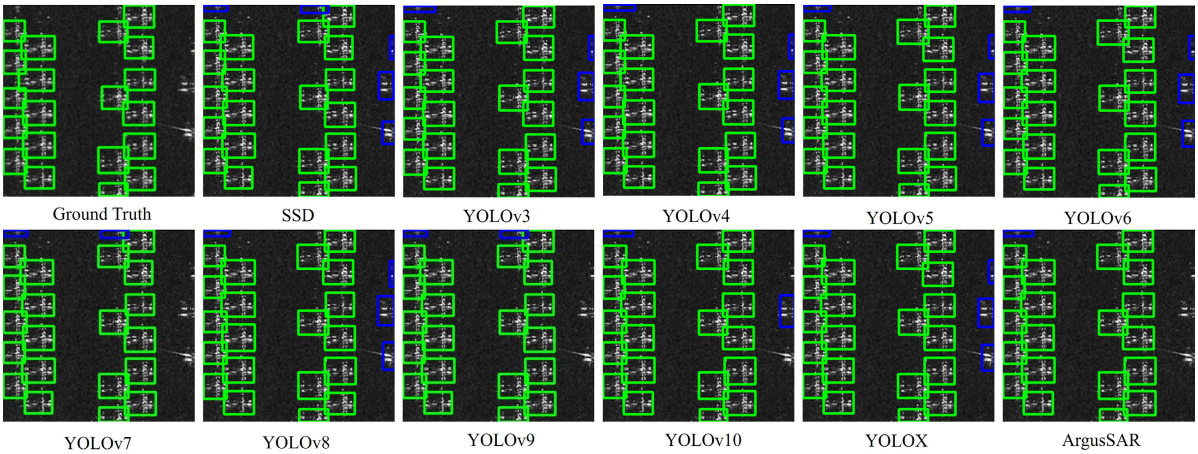


Fig. 9. Detection results on the SADD dataset. In these images, the green boxes, blue boxes, and red boxes represent correct detections, false detections, and missed detections, respectively.

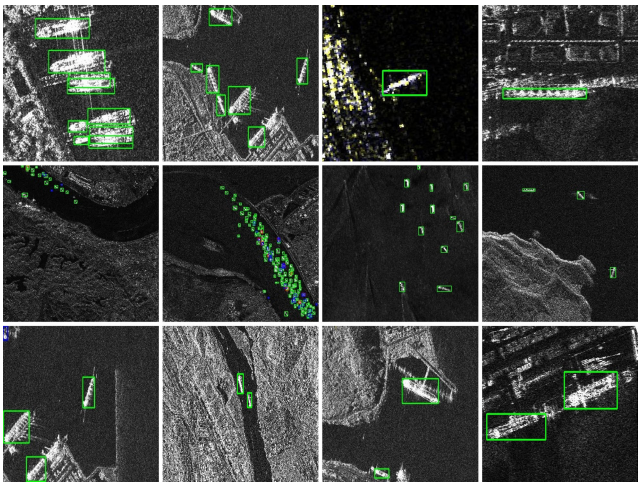


Fig. 10. Detection results for densely arranged ships.

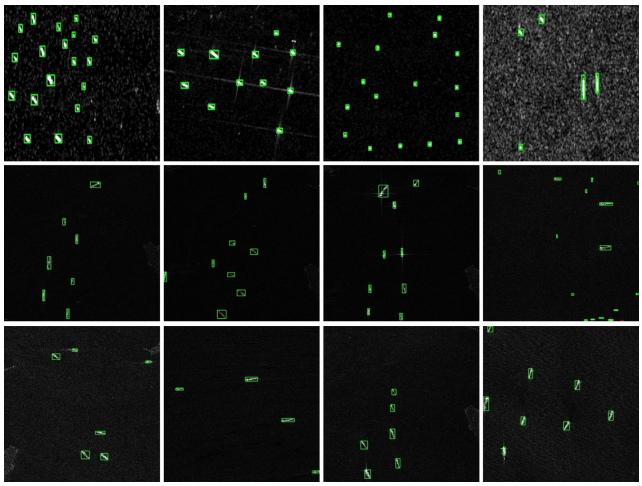


Fig. 11. Detection results for small ship objects.

P and R metrics, which are 2.9% and 1.5% higher than CSD-YOLO, respectively. These experiments confirm that ArgusSAR is more robust when faced with a complex coastal background. The results also reveal that although CSD-YOLO achieves the same 98.6% mAP@.5 as our method, ArgusSAR is significantly superior in both P and R metrics. This indicates

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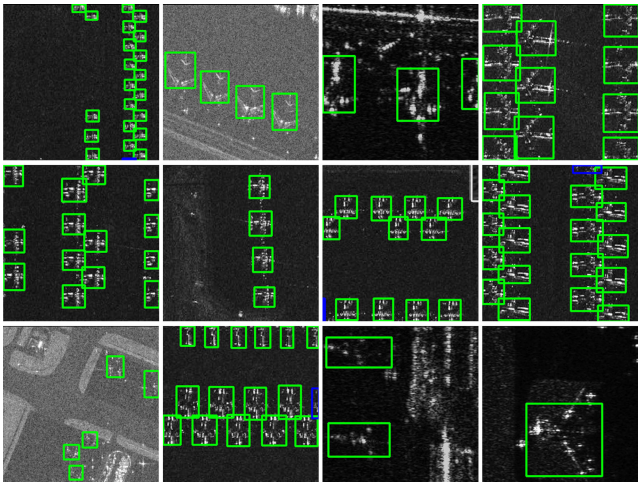


Fig. 12. Detection results for airplane objects.

TABLE III
COMPARISON OF THE DIFFERENT SOTA OBJECT DETECTION METHODS ON THE HRSID DATASET

Method	P (%)	R (%)	F1 (%)	mAP@.5 (%)
YOLO-FA [25]	93.1	87.6	90.3	93.5
CSD-YOLO [28]	93.2	80.4	86.3	86.1
BL-Net [49]	91.6	**89.7**	90.6	88.7
LMSD-YOLO [50]	92.7	86.6	89.5	93.9
HR-SDNet [51]	88.9	82.3	85.5	81.5
FESAR [56]	**93.5**	85.8	89.5	94.0
YOLOv5s-MFC [57]	93.2	89.1	91.1	**94.5**
ARPN [58]	85.7	83.8	84.7	81.8
Quad-FPN [59]	88.0	87.3	87.7	86.1
ArgusSAR	**93.5**	89.4	**91.4**	93.1

our model provides a better overall balance, resulting in fewer false alarms and fewer missed detections. These results demonstrate the SE module's effective suppression of inshore structural interference and the FFAE module's fine-grained feature extraction for small objects.

In Table III, ArgusSAR surpasses the BL-Net by 1.9% and 4.4%, respectively, in the P and mAP@.5 metrics. In addition, the $F1$-score of ArgusSAR is 1.9% higher than that of FESAR, demonstrating that ArgusSAR has a superior performance in detecting different scales of objects. Moreover, although YOLOv5s-MFC obtains a slightly higher mAP@.5 of 94.5%, our ArgusSAR achieves a better balance between P (93.5%) and R (89.4%) compared to 93.2% and 89.1% of YOLOv5s-MFC. This means that while maintaining a high detection rate, our model produces fewer false alarms. This is critical in practical SAR monitoring applications, and we attribute this to the FFAE's precise reconstruction of object boundaries and the SE's effective suppression of land clutter.

In Table IV, ArgusSAR achieves the highest scores across all the metrics, indicating that ArgusSAR also performs well in aircraft detection and can be applied to typical object detection tasks. In detail, although EST-YOLOv5s has a slightly higher P metric, its R metric is 4.4% lower than our ArgusSAR. This implies that EST-YOLOv5s achieves high precision at the cost of numerous missed detections. In contrast, ArgusSAR achieves a much higher recall rate while maintaining almost the same high precision, indicating that our model successfully detects many more small or obscured aircraft objects without introducing additional false alarms.

TABLE IV
COMPARISON OF THE DIFFERENT SOTA OBJECT DETECTION METHODS ON THE SADD DATASET

Method	P (%)	R (%)	F1 (%)	mAP@.5 (%)
EST-YOLOv5s [33]	**97.3**	94.2	95.7	97.8
TPH-YOLOv5 [60]	96.2	70.3	81.2	81.0
SEFEPNet [35]	89.1	98.2	93.4	—
ArgusSAR	97.1	**98.6**	**97.8**	**98.9**

TABLE V
ABLATION STUDY ON THE SSDD DATASET

Experiment	SE	FFAE	MAAD	P (%)	R (%)	F1 (%)	mAP@.5 (%)
A0	×	×	×	93.2	92.3	92.7	96.2
A1	✓	×	×	96.1	96.4	96.2	97.9
A2	×	✓	×	93.6	93.2	93.4	96.3
A3	×	×	✓	94.2	93.2	93.7	97.5
A4	✓	✓	×	97.3	97.1	97.2	98.2
A5	✓	×	✓	97.4	96.8	97.1	**98.8**
A6	×	✓	✓	96.9	97.1	96.9	98.7
A7	✓	✓	✓	**97.8**	**97.4**	**97.6**	98.6

C. Ablation Study

To illustrate the function of the proposed modules, ablation experiments are completed on the SSDD, HRSID, and SADD datasets. Tables V–VII give the results of all possible combinations among the SE, FFAE, and MAAD on the different datasets. The "✓" symbol means that the module is included, and the "×" symbol means that this module is not applied. These results demonstrate that all the modules in our framework are effective and useful.

On the SSDD dataset, in Experiment A1, we use SE, which achieves an improvement with increases of 2.9%, 4%, and 2.7% in P, R, and mAP@.5, respectively. In Experiments A4 and A5, the SE module was, respectively, combined with the FFAE and MAAD modules, showing a significant improvement in the detection abilities. When combining the SE module with the MAAD module, the mAP@.5 increases to 98.8%. In detail, the feature maps processed through the attention module contain a substantial amount of weighted spatial information, and the MAAD module can strengthen awareness of the spatial position by fully using the information of the feature map. In Experiment A6, we did not use the SE module, which means that the interaction between channels in the feature maps cannot be considered, resulting in a lower detection precision when compared to Experiments A4 and A5. Experiments A2, A4, and A6 indicate that using the FFAE module can significantly improve the R. This is due to the FFAE module considering the impact of other neighboring pixels and using the weighted feature maps, which obtain abundant detail information and avoid discontinuities in grayscale values after sampling. According to Experiment A7, it is worth mentioning that combining all three modules can, respectively, improve the detection performance, with the

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TABLE VI
ABLATION STUDY ON THE HRSID DATASET

Experiment	SE	FFAE	MAAD	P (%)	R (%)	F1 (%)	mAP@.5 (%)
B0	×	×	×	88.4	86.7	87.5	90.2
B1	✓	×	×	92.9	**90.1**	**91.5**	92.7
B2	×	✓	×	89.6	72.7	80.3	79.8
B3	×	×	✓	89.3	73.8	80.8	78.2
B4	✓	✓	×	**94.1**	86.3	90.0	**93.2**
B5	✓	×	✓	93.1	87.7	90.3	91.5
B6	×	✓	✓	90.7	87.0	88.8	91.6
B7	✓	✓	✓	93.5	89.4	91.4	93.1

TABLE VII
ABLATION STUDY ON THE SADD DATASET

Experiment	SE	FFAE	MAAD	P (%)	R (%)	F1 (%)	mAP@.5 (%)
C0	×	×	×	92.1	91.8	91.9	93.7
C1	✓	×	×	94.7	94.5	94.6	97.9
C2	×	✓	×	96.3	93.4	94.8	96.9
C3	×	×	✓	89.4	73.5	80.7	78.4
C4	✓	✓	×	**97.4**	98.4	**97.9**	98.7
C5	✓	×	✓	94.2	96.7	95.4	96.2
C6	×	✓	✓	95.1	95.7	95.4	97.3
C7	✓	✓	✓	97.1	**98.6**	97.8	**98.9**

P increasing by 5.1%, the R increasing by 5.1%, and the mAP@.5 increasing by 2.4%.

On the HRSID dataset, the experimental results demonstrate that each module contributes significantly to improving detection performance in Table VI. Experiment B1, which includes only the SE module, shows an increase in P, R, and mAP@.5 by 4.5%, 3.4%, and 2.5%, respectively. Combining SE with FFAE in Experiment B4 leads to a substantial improvement in R, highlighting the ability to capture detailed information from neighboring pixels of FFAE. However, lacking SE in Experiment B6 results in lower detection accuracy due to the lack of channel interaction. Experiments B2, B4, and B6 confirm that FFAE significantly improves R by considering weighted feature maps. Finally, integrating all three modules in Experiment B7 achieves the best performance, with P increasing by 4.8%, R by 2.7%, and mAP@.5 by 2.9%.

The ablation study on the SADD dataset further validates the contributions of the SE, FFAE, and MAAD modules. In Table VII, Experiment C1, utilizing only the SE module, demonstrates improvements in P, R, and mAP@.5 by 2.6%, 2.7%, and 4.2%, respectively. The combination of SE and FFAE in Experiment C4 enhances R, showing the ability of FFAE in capturing fine-grained details. Experiments C2, C4, and C6 highlight the significant impact of FFAE on R by using weighted feature maps. This strategy ensures effective feature aggregation and superior detection performance across diverse scenarios.

To confirm the detection ability and practical application value of ArgusSAR, Fig. 13 shows a visualized attention heatmap. The region where the color is close to red demonstrates that the detector focuses more on this region when the detector is working. It is clear that ArgusSAR pays more attention to the ship and aircraft objects, while the baseline is easily affected by the interference facilities. This reflects that ArgusSAR can accurately detect objects in complex backgrounds, so that the detector can focus more on the objects of interest rather than the land.

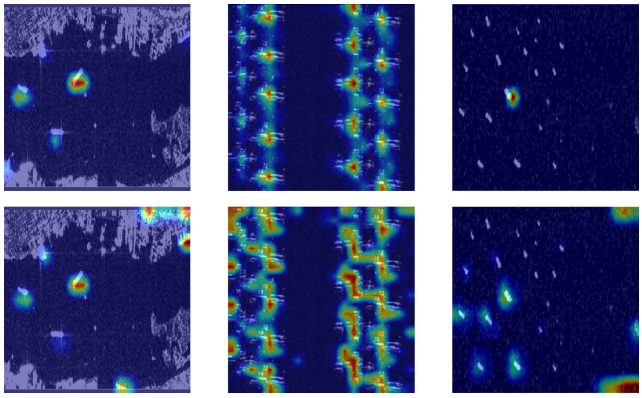


Fig. 13. Visualized attention heatmaps for detectors visualized. First row: result of ArgusSAR. Second row: result of the YOLOv10.

TABLE VIII
COMPARISON OF DIFFERENT ATTENTION MODULES

Attention module	P (%)	R (%)	F1 (%)	mAP@.5 (%)
/	93.2	92.3	92.7	96.2
ECA	95.7	95.4	95.5	97.5
CA	94.8	89.3	92.0	94.5
CBAM	91.8	94.2	93.0	96.4
SimAM	95.2	**96.7**	96.0	97.6
GAM	94.3	90.9	92.6	96.1
SA	93.0	91.4	92.2	95.9
MHA	**96.1**	96.4	**96.2**	**97.9**

D. Discussion

1) Comparison of Attention Modules: As different attention modules have different influences on the detection performance, we conducted comparative experiments with seven attention modules: MHA, CA, ECA, CBAM, simple and parameter-free attention mechanism (SimAM), global attention mechanism (GAM), and shuffle attention (SA). Table VIII lists the quantitative evaluation results on the SSDD dataset.

It is clear that not all the attention modules can enhance the performance, especially the SA module, which weakens the detection performance. Some attention modules result in an unbalanced impact on the different evaluation metrics. In detail, the CA and GAM attention modules improve P by 1.6% and 1.1%, respectively, but both the R and mAP@.5 are clearly reduced. In addition, the CBAM attention module improves the R by 1.9%, but the P is lower than that of the baseline. Furthermore, the MHA obtains a higher P than the SimAM attention module, but SinAM surpasses SE by 1.3% in R, and shows the best improvement in R. Moreover, the MHA module improves the P, R, and mAP@.5 by 2.9%, 4.1%, and 1.7%, respectively, and obtains the best improvement. Thus, we chose the MHA to enhance the proposed detector.

2) Parameter Selection for Upsampling Operator: By adjusting the k_{up} and $k_{encoder}$ parameters, the performance of the upsampling operator can be changed. To obtain the best performance from SE, we chose different combinations

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TABLE IX
RESULTS OF DIFFERENT PARAMETER COMBINATIONS k_{UP} AND $k_{ENCODER}$

Parameter		SSDD (Ship Dataset)				HRSID (Ship Dataset)				SADD (Aircraft Dataset)			
$k_{encoder}$	k_{up}	P (%)	R (%)	F1 (%)	mAP@.5 (%)	P (%)	R (%)	F1 (%)	mAP@.5 (%)	P (%)	R (%)	F1 (%)	mAP@.5 (%)
—	—	93.2	92.3	92.7	96.2	88.4	86.7	87.5	90.2	92.1	91.8	91.9	93.7
1	3	92.5	92.0	92.2	96.4	89.1	71.9	79.6	78.6	91.9	92.6	92.2	93.9
1	5	92.5	93.4	92.9	96.5	83.2	74.5	78.6	76.4	92.6	89.9	91.2	93.1
1	7	92.9	93.0	92.9	95.9	84.3	72.4	77.9	80.7	93.4	92.6	93.0	94.5
1	9	92.6	91.6	92.1	95.8	77.1	70.2	73.5	76.9	92.1	93.8	92.9	94.4
3	3	91.4	89.7	90.5	94.9	88.2	73.1	80.0	**81.3**	93.7	92.4	93.0	95.2
3	5	94.8	88.8	91.7	96.1	89.3	73.2	80.5	79.9	95.3	**93.6**	94.4	96.4
3	7	**93.6**	**93.2**	**93.4**	**96.3**	**89.6**	72.7	80.3	79.8	**96.3**	93.4	**94.8**	**96.9**
3	9	93.9	91.7	92.8	96.7	88.6	70.9	78.8	77.2	95.8	91.2	93.4	96.1
5	7	91.9	93.0	92.4	95.3	88.9	**73.7**	**80.6**	78.4	96.1	92.7	94.4	96.7
5	9	93.4	90.1	91.7	97.1	87.2	72.5	79.2	79.6	94.6	92.5	93.5	95.9
7	9	94.3	88.7	91.4	95.5	89.1	72.7	80.1	79.7	95.3	92.1	93.7	95.7

TABLE X
COMPARISON OF THE NUMBER OF PERCEPTION LAYERS

Number of layers	P (%)	R (%)	F1 (%)	mAP@.5 (%)
—	93.2	92.3	92.7	96.2
1	93.4	92.9	93.1	96.6
2	**94.2**	**93.2**	**93.7**	**97.5**
3	95.1	90.6	92.8	96.5
4	94.5	91.1	92.8	96.9
5	91.3	90.2	90.7	95.8
6	91.5	89.7	90.6	94.3

TABLE XI
PERFORMANCE COMPARISON OF DIFFERENT FEATURE EXTRACTION MODULES ON THE HRSID DATASET

Extraction Method	mAP@.5 (%)	AP_s (%)	AP_m (%)	AP_l (%)
MobileNet [61]	76.9	53.1	52.2	22.8
EfficientNet [62]	84.3	63.2	57.0	29.3
SE (Ours)	**93.1**	**70.9**	**68.4**	**36.7**

of parameters of k_{up} and $k_{encoder}$ on the SSDD, HRSID, and SADD datasets, which are shown in Table IX.

From Table IX, some combinations of parameters are not helpful to the performance of the detector and can even cause a reduced detection effect. When the parameters are $k_{encoder} = 3$ and $k_{up} = 7$, the upsampling module brings significant improvements to all the evaluation metrics across the SSDD, HRSID, and SADD datasets. Thus, we choose $k_{encoder} = 3$ and $k_{up} = 7$ as the combination of parameters.

3) Number of MAAD Perception Layers: We evaluated the efficiency of the MAAD by selecting different numbers of layers. In Table X, the different numbers of MAAD layers (1–6 blocks) are chosen and compared the detection performance with the baseline on SSDD dataset.

From Table X, it is worth noting that two MAAD layers achieved optimal detection performance, which improves the P, R, and mAP@.5 metrics by 1.0%, 0.9%, and 1.3%, respectively. In addition, with the increase in blocks, the mAP@.5 metric is significantly decreased compared to the baseline. Besides, redundant blocks even cause a reduction in detection performance. Thus, we set the number of MAAD layers to 2.

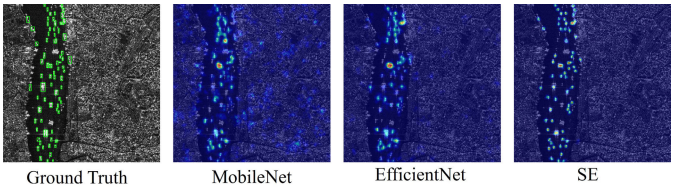


Fig. 14. Visualization of feature maps generated by different feature extraction methods. Original SAR images with ground-truth boxes. Feature map heatmaps (MobileNet, EfficientNet, and SE). Warmer colors (red/yellow) indicate stronger feature activation, while cooler colors (blue) represent lower activation.

TABLE XII
PERFORMANCE COMPARISON OF DIFFERENT FEATURE FUSION MODULES ON THE HRSID DATASET

Fusion Method	mAP@.5 (%)	AP_s (%)	AP_m (%)	AP_l (%)
FPN [63]	80.2	57.2	54.9	19.5
PANet [64]	87.5	67.3	66.0	25.8
BiFPN [65]	90.6	68.5	67.1	29.2
ASFF [66]	91.8	69.3	66.9	31.1
FFAE (Ours)	**93.1**	**70.9**	**68.4**	**36.7**

4) Comparison of Feature Extraction Methods: To further validate the superiority of our proposed SE module, we selected ArgusSAR as the framework and kept the FFAE and detection head MAAD identical while only replacing the feature extraction module. Our SE module demonstrates significant performance advantages over the lightweight backbones MobileNet and EfficientNet when integrated into the ArgusSAR framework. As shown in Table XI, SE achieves an mAP@.5 of 93.1% on the HRSID dataset, substantially outperforming EfficientNet (84.3%) and MobileNet (76.9%). Furthermore, the superiority is shown in small object detection (AP_s), where SE (70.9%) surpasses EfficientNet (63.2%) and MobileNet (53.1%). This highlights that the SE module can enhance the ability to extract discriminative features for small objects in SAR images. The qualitative results in Fig. 14 show that the feature maps generated by the SE module are more precisely concentrated on the actual object locations compared to those from MobileNet and EfficientNet.

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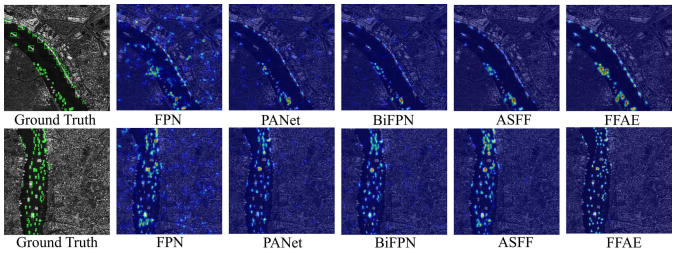


Fig. 15. Visualization of feature maps generated by different feature fusion methods. The first column shows the original SAR images with ground-truth boxes. The subsequent columns (FPN, PANet, BiFPN, ASFF, and MAAD) display the corresponding feature map heatmaps. Warmer colors (red/yellow) indicate stronger feature activation, while cooler colors (blue) represent lower activation.

5) Comparison of Feature Fusion Methods: As the critical module between the backbone and the detection head, the feature fusion module directly impacts the model's ability to detect multiscale objects. To further validate the superiority of our proposed FFAE, we selected ArgusSAR as the framework and kept the backbone SE and detection head MAAD identical while only replacing the feature fusion module. We chose FPN [63] and PANet [64] to represent the classic baseline based on a path-enhancement method for multiscale fusion. We introduced BiFPN [65] and ASFF [66] to represent efficient weighted fusion methods and an adaptive spatial fusion method, respectively. These four strategies together formed a comprehensive and highly challenging performance benchmark. All models were trained and evaluated on the HRSID dataset under identical settings, and the results are shown in Table XII. Quantitatively, our method achieves 93.1% mAP@.5 on the HRSID dataset, significantly outperforming all baseline feature fusion strategies, including the advanced ASFF (91.8%) and BiFPN (90.6%). The advantage of the scale-specific metrics is that our model attains an AP_s of 70.9%, demonstrating a remarkable advantage over all methods in the challenging task of small object detection, and our method also achieves the highest scores for medium (AP_m) and large (AP_l) objects.

To show the differences in feature representation among the modules, we visualize the feature maps generated by different methods for the same SAR image containing dense, small objects in Fig. 15. In these challenging scenes, traditional methods, like FPN and PANet, produce messy and diffuse feature maps. They fail to separate closely arranged objects and miss the small objects almost entirely in the bottom-row example. Although BiFPN and ASFF show improvement, their feature maps still contain some false detections and missed detections. In contrast, our method generates bright, highly focused, and well-separated activation peaks for each object. Even in the extremely dense, small-object scene (bottom row), our module successfully highlights each object with a distinct and strong signal. This visualization confirms that our FFAE module produces cleaner and higher-fidelity feature representations that are robust to background noise and highly sensitive to objects of all scales.

In the HRSID dataset, the detection accuracy for large ships (AP_l) is lower than that for small ships (AP_s), which is similar to the ELLK-Net [31]. The lower AP_l indicates that the environments for large objects in the HRSID dataset have a much lower signal-to-noise ratio and significantly higher complexity than the environments for small objects. This phenomenon stems from the dataset's specific scene distribution. Large ships tend to be located in inland complex interference scenes, which contain significant land noise and strong clutter. As a result, models struggle to accurately separate large objects from the complex background. In contrast, many small ship objects are often located in areas with relatively clean backgrounds. The primary detection challenge for them is their small size, not severe background confusion.

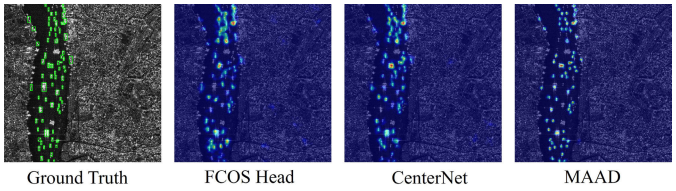


Fig. 16. Visualization of feature maps generated by different feature extraction methods. Original SAR images with ground-truth boxes. Feature map heatmaps (FCOS Head, CenterNet Head, and MAAD). Warmer colors (red/yellow) indicate stronger feature activation, while cooler colors (blue) represent lower activation.

TABLE XIII
PERFORMANCE COMPARISON OF DIFFERENT DETECTION HEADS ON THE HRSID DATASET

Detection Head	mAP@.5 (%)	AP_s (%)	AP_m (%)	AP_l (%)
FCOS Head [67]	87.7	65.4	63.6	**38.3**
CenterNet Head [68]	88.9	67.2	68.1	30.5
MAAD (Ours)	**93.1**	**70.9**	**68.4**	36.7

6) Comparison of Detection Head: To further validate the superiority of our proposed MAAD module, we selected ArgusSAR as the framework and kept the backbone SE and the FFAE identical while only replacing the detection head.

In Table XIII, the comparison between our MAAD and other popular detection heads validates MAAD's effectiveness within the ArgusSAR architecture. The results reveal that MAAD achieves the highest mAP@.5 (93.1%) on the HRSID dataset, clearly outperforming both FCOS Head (87.7%) and CenterNet Head (88.9%). Notably, MAAD also leads in small and middle object detection, especially in AP_s with a score of 70.9%, compared to 65.4% for FCOS and 67.2% for CenterNet. The feature map visualizations in Fig. 16 claim that MAAD generates clear and intensely focused feature maps on the objects compared to FCOS Head and CenterNet Head. This indicates that MAAD's multidimensional attention mechanism successfully refines the features, enhancing the localization signal and improving the detector's ability to distinguish multiscale objects from the background clutter.

7) Limitations and Future Work: ArgusSAR has demonstrated strong robustness and generalization across complex scenarios, such as strong speckle noise, high-clutter backgrounds, and densely distributed objects, in evaluations on the SSDD, HRSID, and SADD datasets. Nevertheless, our work is focused on single-class detection, and its efficacy

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in multiclass scenarios remains unexplored. Furthermore, the physical mechanism is not explicitly exploited to constrain our method, which may limit further performance gains. In future work, the ArgusSAR model will be applied to more complex and diverse multiclass detection scenarios. In parallel, we will focus on optimizing our model by incorporating the physical mechanisms of SAR imaging, thereby enhancing its generalization capability and detection stability.

IV. Conclusion

To solve the SAR object detection problems of complex inference, different scales, and weak robustness, ArgusSAR, which is a robust SAR object detector for ship and aircraft objects, has been proposed. Experiments on the SSDD, HRSID, and SADD datasets demonstrate the effectiveness and robustness of ArgusSAR.

1) The ArgusSAR framework obtains the best performance in three different datasets with ship and aircraft. The P metric reaches 97.8% and the mAP@.5 metric reaches 98.6% on the SSDD dataset, which represents an SOTA performance in SAR images. Moreover, experiments on the HRSID dataset show the excellent robustness of ArgusSAR for multiscale ship detection, which obtains the highest accuracy among the other detectors. Besides, ArgusSAR obtains a superior performance in aircraft object detection on the SADD dataset. Thus, the proposed method can be better used for practical detection tasks.

2) The rationality and effectiveness of the designed modules in ArgusSAR are verified in ablation experiments on the SSDD, HRSID, and SADD datasets. In detail, to suppress the interference in SAR images, we propose SE to weigh the different channels of the feature maps, which improves the $F1$-score by 3.5%, 4%, and 2.7% on the three different datasets, respectively.

3) Due to the considerable lack of information caused by nearest neighbor upsampling, we design an FFAE to extract feature maps with more boundary and semantic information. This method obtains high R metrics on the three different datasets, which verifies that ArgusSAR can preserve the boundary information of different layers.

4) To enhance the detector's generalization across different SAR datasets, we introduce the MAAD to construct output channels of multiscale and multitask perception with the self-attention. By enhancing the perception of scale, spatial location, and task, the decoder improves the generalization of the detector by fully using the information of the feature maps, which results in improved performance.

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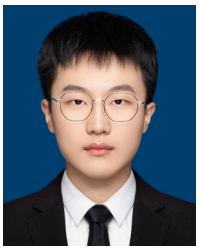
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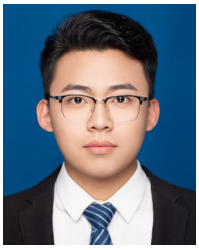
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